

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 June 2026
Team ID	
Project Name	ShopEZ – MERN Stack E-Commerce Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Create account with name, email, password
FR-2	User Authentication	Secure login/logout using JWT
FR-3	Product Catalog	Display all available products
FR-4	Product Search	Users can search products by name.
FR-5	Category Management	Browse products category-wise
FR-6	Shopping Cart	Add products to cart
FR-7	Shopping Cart	Increase product quantity
FR-8	Shopping Cart	Decrease product quantity
FR-9	Shopping Cart	Remove products from cart
FR-10	Order Management	Place orders from cart
FR-11	Product Images	Display product images
FR-12	Admin Dashboard	Add new products
FR-13	Admin Dashboard	Update existing products
FR-14	Admin Dashboard	Delete products
FR-15	Admin Dashboard	Manage product inventory and categories

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	User-friendly interface with easy navigation and responsive design
NFR-2	Security	Secure authentication using JWT and protected routes
NFR-3	Reliability	Consistent access to products, cart, and order services
NFR-4	Performance	Fast loading of product listings and search results
NFR-5	Availability	System available to users 24/7 with minimal downtime